



MANIPAL
UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)



INTERNATIONAL INNOVATION CHALLENGE 3.0



Theme: Aerotech & Aerospace



Idea Title: FIRST LIGHT



Team ALPHA

IDEA TITLE

- ✓ PCC: AI maneuvers proven safe before executing
- ✓ 5-agent pipeline, fully automated
- ✓ No blind trust — every command needs proof
- ✓ Live audit trail + adversarial Attack Library

TECHNICAL APPROACH



Technologies

FastAPI + SQLite backend, Claude (Anthropic) for mission planning, Z3 SMT solver for formal proofs, WebSocket real-time telemetry, SGP4 orbit propagation, browser-based mission control UI

Methodology

Planner → Dynamics → Safety → Proof Generator
→ Reviewer pipeline; every step's latency, confidence, and reasoning is genuinely computed and hash-chained into a tamper-evident audit trail

FEASIBILITY AND VIABILITY



Feasibility

Working end-to-end prototype:
real LLM calls, real Z3 proofs,
real digital-twin telemetry — 157
automated tests passing against
the live backend



Challenges

LLM proposal latency (~3s)
vs. real-time flight timing;
integrating with certified flight
software (NASA cFS)



Strategies

Deterministic fallback verifier
decoupled from LLM latency;
phased NASA cFS integration via
the existing interface layer; proof
caching for repeat maneuvers

IMPACT AND BENEFITS

- 🎯 Safer autonomy for CubeSats & small-sats
- 🌿 Safety, reusable science, lower mission risk

RESEARCH AND REFERENCES

AFRL STARS Runtime Assurance for spacecraft attitude control — Black et al., arXiv:2402.14723
Watchdogs and Oracles: runtime verification for LLM-driven autonomy — arXiv:2511.14435
Glass Box at Orbit: constitutional AI verification for CubeSat autonomy — arXiv:2606.02967
NASA Core Flight System (cFS) — open-source flight software reference architecture, cfs.gsfc.nasa.gov